EXP 3

EDA – Data Import and Export

AIM:

To import data into a pandas DataFrame from different sources, namely a CSV file, an Excel file, an SQL (SQLite) database and a web page (web scraping), and to understand how different data formats are handled.

ALGORITHM:

1. Import the libraries pandas, sqlite3, requests, BeautifulSoup and StringIO.
2. Read the Titanic CSV file using read_csv() and display the first five rows.
3. Read the heart statlog Excel file using read_excel() and display the first five rows.
4. Connect to an SQLite database (employee.db) and create a cursor.
5. Create the table employee2 with the columns id, name, department, salary and hire_date.
6. Insert five employee records into the table using executemany() and save them using commit().
7. Read the table into a DataFrame using read_sql() with a SELECT query and display it.
8. Close the database connection.
9. Send a request to the Worldometers web page using requests.get() and check the status code.
10. Parse the page using BeautifulSoup and find all the tables on the page.
11. Convert the first table into a DataFrame using read_html() and display the first five rows.

CODE:

```
import pandas as pd
import sqlite3
import requests
from bs4 import BeautifulSoup
from io import StringIO
```

```
csv_df = pd.read_csv("Titanic-Dataset.csv")
print("CSV Data")
print(csv_df.head())
```

```
excel_df = pd.read_excel("heart stalog dataset.xlsx")
print("Excel Data")
print(excel_df.head())
```

```
conn = sqlite3.connect("employee.db")

cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS employee2(
    id INTEGER,
    name TEXT,
```

```
    department TEXT,
    salary REAL,
    hire_date TEXT
)
""")

conn.commit()
```

```
employees = [
    (1,"Alice Smith","HR",55000,"2018-05-01"),
    (2,"Bob Johnson","IT",72000,"2019-07-15"),
    (3,"Carol White","Finance",68000,"2017-09-30"),
    (4,"David Brown","Marketing",60000,"2020-02-10"),
    (5,"Eva Green","IT",75000,"2021-04-25")
]

cursor.executemany("""
INSERT INTO employee2
VALUES (?,?,?,?,?)
""", employees)

conn.commit()
```

```
sql_df = pd.read_sql("SELECT * FROM employee2", conn)

print("SQL Data")
print(sql_df.head())
```

```
conn.close()
```

```
url = "https://www.worldometers.info/geography/countries-of-the-world/"
response = requests.get(url)

print("Status Code:", response.status_code)

soup = BeautifulSoup(response.text, "html.parser")
```

```
tables = soup.find_all("table")

print("Total Tables:", len(tables))
```

```
web_df = pd.read_html(StringIO(str(tables[0])))[0]

print("Web Scraped Data")
print(web_df.head())
```

OUTPUT:

Output of: csv_df = pd.read_csv("Titanic-Dataset.csv")

```
CSV Data
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
```

```
4            5         0       3

                                                Name      Sex   Age  SibSp  \
 0                             Braund, Mr. Owen Harris     male  22.0      1
 1  Cumings, Mrs. John Bradley (Florence Briggs Th...   female  38.0      1
 2                              Heikkinen, Miss. Laina   female  26.0      0
 3        Futrelle, Mrs. Jacques Heath (Lily May Peel)   female  35.0      1
4                            Allen, Mr. William Henry     male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

Output of: excel_df = pd.read_excel("heart stalog dataset.xlsx")

```
Excel Data
   age  sex  chest  resting_blood_pressure  serum_cholestoral  \
0   70    1      4                     130                322
1   67    0      3                     115                564
2   57    1      2                     124                261
3   64    1      4                     128                263
4   74    0      2                     120                269

   fasting_blood_sugar  resting_electrocardiographic_results  \
0                    0                                     2
1                    0                                     2
2                    0                                     0
3                    0                                     0
4                    0                                     2

   maximum_heart_rate_achieved  exercise_induced_angina  oldpeak  slope  \
0                          109                        0      2.4      2
1                          160                        0      1.6      2
2                          141                        0      0.3      1
3                          105                        1      0.2      2
4                          121                        1      0.2      1

   number_of_major_vessels  thal     class
0                        3     3   present
1                        0     7    absent
2                        0     7   present
3                        1     7    absent
4                        1     3    absent
```

Output of: sql_df = pd.read_sql("SELECT * FROM employee2", conn)

```
SQL Data
   id         name department    salary   hire_date
0   1  Alice Smith         HR   55000.0  2018-05-01
1   2  Bob Johnson         IT   72000.0  2019-07-15
2   3  Carol White    Finance   68000.0  2017-09-30
3   4  David Brown  Marketing   60000.0  2020-02-10
4   5    Eva Green         IT   75000.0  2021-04-25
```

Output of: url = "https://www.worldometers.info/geography/countries-of-the-world/"

```
Status Code: 200
```

Output of: tables = soup.find_all("table")

```
Total Tables: 2
```

Output of: web_df = pd.read_html(StringIO(str(tables[0])))[0]

```
Web Scraped Data
   #         Country  Population 2026             Region
0  1           India       1476625576               Asia
1  2           China       1412914089               Asia
2  3  United States        349035494  Northern America
3  4       Indonesia        287886782               Asia
4  5        Pakistan        259299791               Asia
```

RESULT:

Thus, data was imported into pandas DataFrames from a CSV file, an Excel file, an SQLite database table (five employee records) and an HTML table scraped from a web page (status code 200, 2 tables found), and the first rows of each were displayed.